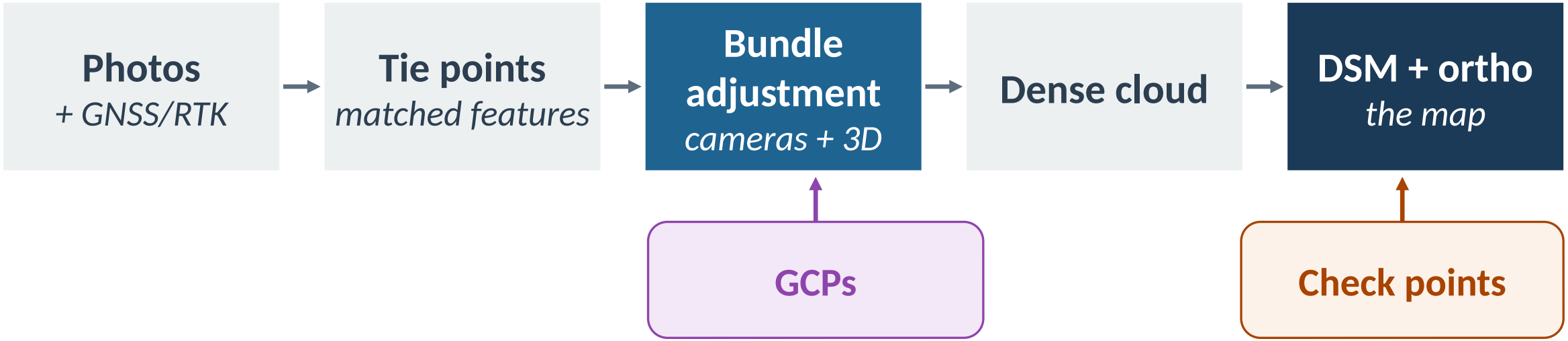
Drone mapping

Ground control, photos for structure from motion, mission planning



From photos to a map



- **Today:** the input – control points, photos, flight plan
- **After the flights:** WebODM, orthomosaic, DSM, accuracy

Further reading: Photogrammetry · Structure from motion · Bundle adjustment · Orthophoto · Digital elevation model

Today

- **Control:** ground control points and check points, targets
- **Photos:** overlap, pixel size, sharpness, light
- **Plan:** from flying height to flight lines and flight time
- **Then, on the field day:** panels out, RTK, flights



1 Control

Ground control points and check points

Photos → map | **Control** | Photos | Plan | Results



Ground control and check points

Ground control point (GCP)

- Known coordinates, visible in the photos
- **Used** in the bundle adjustment
- Fixes position, scale, rotation, height

Check point

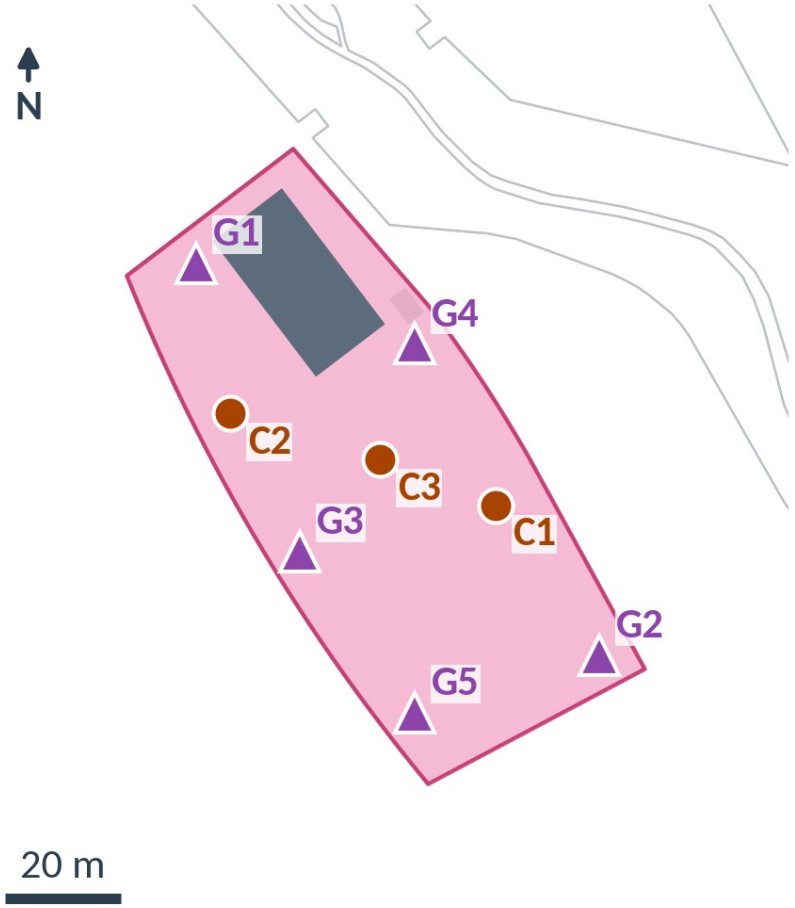
- Measured the same way
- **Not used** in the processing
- Tests the result: error in E, N, H

Pause: why can a GCP not tell you how good your map is?

Further reading: Ground control point · Accuracy and precision · Root mean square deviation



How many, and where?



- **≥ 5 GCPs**: edges and middle
- Each one in ≥ 5 photos
- **Check points** in between, never used as GCPs
- Open sky, flat ground, away from walls
- Final places: decided in the field

Further reading: Ground control point · Bundle adjustment

Map: ALKIS (simplified WFS) and height benchmarks: Geobasis NRW, dl-de/zero-2-0; layout: R. Becker; rules: OpenDroneMap documentation (GCP)

Georeferencing: two questions

corrections from ↓	precise points on the ground	precise camera positions
SAPOS	GCPs + RTK (Emlid) **cm**	drone RTK (mobile data) or PPK **cm**
own base (Emlid)	GCPs, base + rover **cm relative***	drone RTK or PPK from own base **cm relative***
nothing / maps	virtual GCPs: DOP + DGM1 **dm**	drone GNSS only **m**

Every cell: check points · * absolute only if the base position is known

Further reading: Real-time kinematic positioning · Ground control point · Orthophoto



Think first

What must my map fit?

my map must …	method	fit
count trees, area of a field	?	
crop change since the last flight	?	
overlay with DOP, ALS, cadastre	?	
volume of a pile, accuracy study	?	

Pause: pick a method from the grid for each task – and why

Further reading: Image registration · Accuracy and precision



Solution

What must my map fit?

my map must …	method	fit
count trees, area of a field	drone GNSS only	m
crop change since the last flight	co-register to the first flight	relative
overlay with DOP, ALS, cadastre	virtual GCPs or drone RTK	dm
volume of a pile, accuracy study	RTK/PPK + GCPs + check points	cm

Cheapest method that meets the purpose – proven with check points

Further reading: Image registration · Accuracy and precision



Our choice at the Green Fab Lab

- **Ground:** 5 GCPs + 3 check points: SAPOS and our own base
- **Camera:** one mission, three times: SAPOS / own base / none + PPK
- **After the flights:** each with and without GCPs – errors on the check points
- **You find out yourselves which cell is worth the effort**

Further reading: Real-time kinematic positioning



Corrections and heights: what you need from geo0260

- **Base on a known point → corrections:** DGNSS dm, RTK cm (FIX)
- **SAPOS HEPS:** network RTK, needs mobile data
- **Own base:** LoRa to the rover, NTRIP to the drone, or PPK
- **Heights:** GNSS h ≈ NHN + 45 m here
- **One height system** for photos, GCPs, check points

Details: slides geo0260 GNSS, RTK and SAPOS (parts 2–5)



Which method when?

	SAPOS + mobile data	SAPOS, dead spot	no network, mobile data	no network, no mobile data
SAPOS network RTK	standard	–	–	–
own base, known mark	rarely	rover by LoRa, drone PPK	standard [1]	rover by LoRa, drone PPK [1]
PPK (drone + base log)	fallback	standard	fallback [1]	standard [1]
drone GNSS only	m-level tasks	m-level tasks	m-level tasks	m-level tasks

[1] mark measured once: long static log + PPP

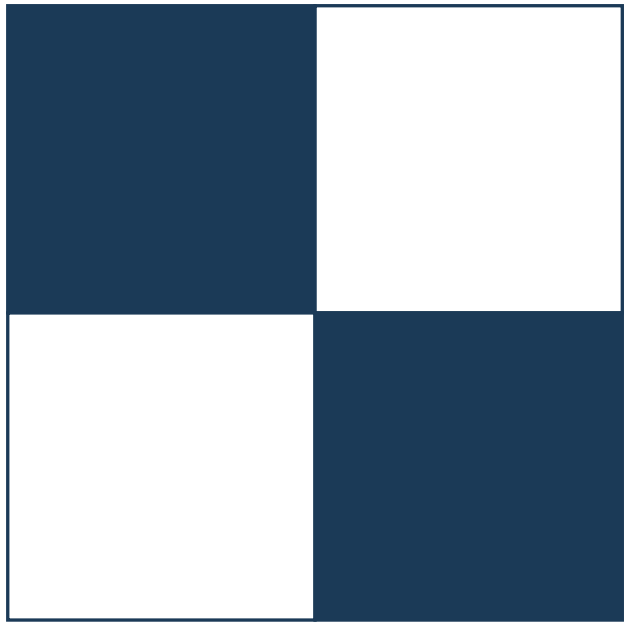
Always: check points



Targets you can find in the photos

50 cm panel	RGB	MS
at 25 m	73 px	44 px
at 60 m	31 px	18 px

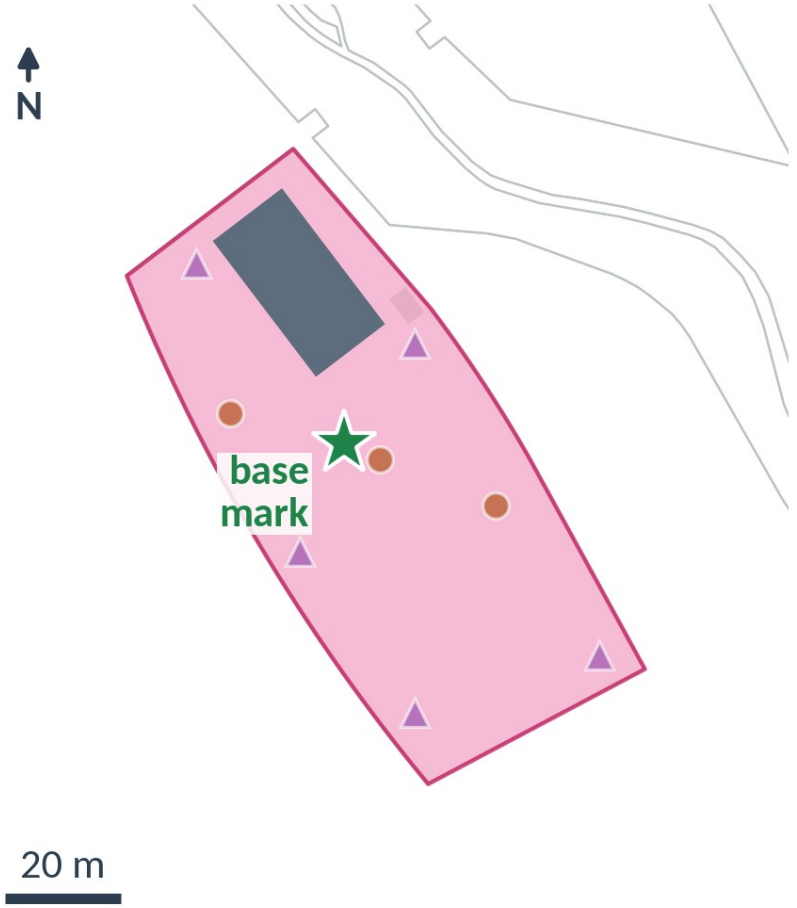
- Measure the centre: FIX, ≈ 30 s
- Measure again later
- Photo of each panel + name
- Leave it until the drone has landed



matte, flat, weighted

Further reading: Fiducial marker · Ground sample distance

Our base mark at the Green Fab Lab



- **No official mark nearby** (all destroyed) → our own
- Nail in a paving joint, south of the building
- **Measure:** SAPOS RTK, 2×, ≥ 1 h apart
- **Or check:** 2 h static log + PPK
- Photo, sketch, E N h H
- **Field day:** set it, measure it, base on it
- **Then:** rover and drone with our own corrections

Further reading: Survey marker · Tribrach

Map: ALKIS (simplified WFS) and height benchmarks: Geobasis NRW, dl-de/zero-2-0; mark status: TIM-online; base mark: planned (R. Becker)

GCPs for WebODM

```
EPSG:25832
329851.42 5708012.37 26.91 2734 1880 DJI_0123.JPG G1
329851.42 5708012.37 26.91 1022 2410 DJI_0124.JPG G1
329893.10 5707990.85 27.05 3911  760 DJI_0157.JPG G2
```

- **Per line:** E N H, pixel x y, photo, name
- Pixels: click in the WebODM GCP interface
- **One height system:** NHN vs. ellipsoid ≈ 45 m
- **Photo heights:** h or EGM96? Check over the base mark

Format: OpenDroneMap documentation, Ground Control Points (docs.opendronemap.org/gcp)



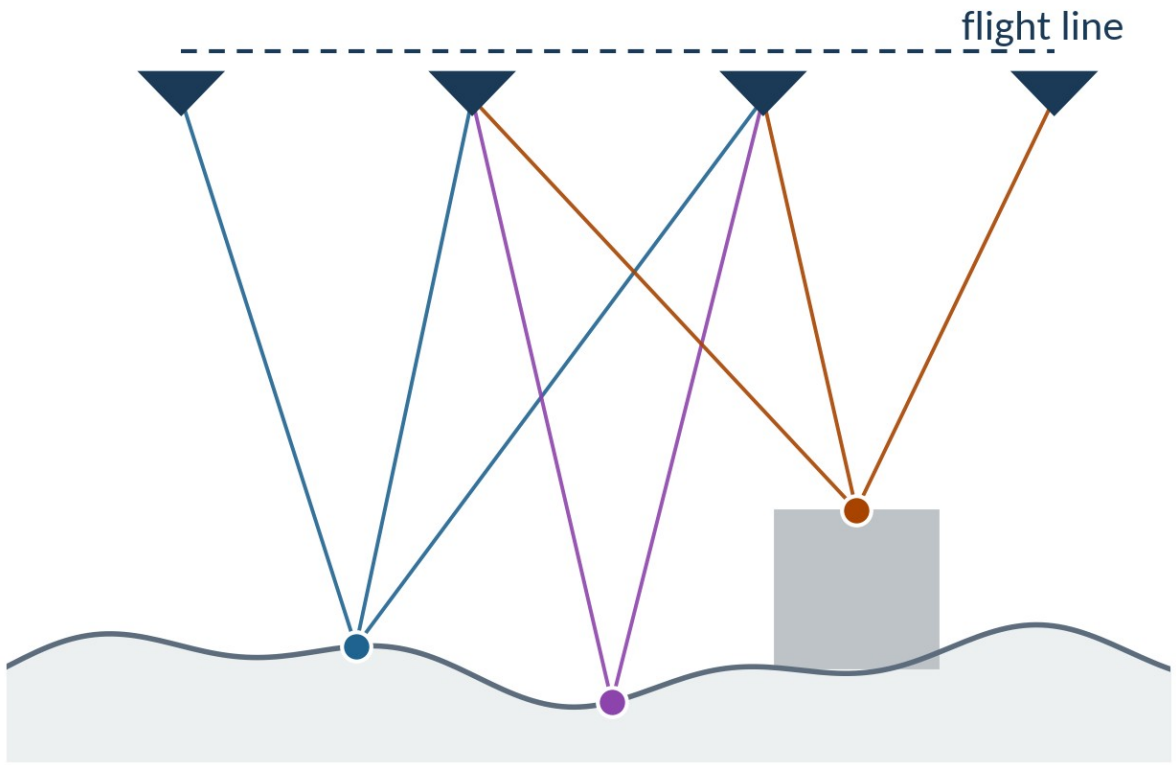
2 Photos

What structure from motion needs

Photos → map | Control | **Photos** | Plan | Results



Structure from motion

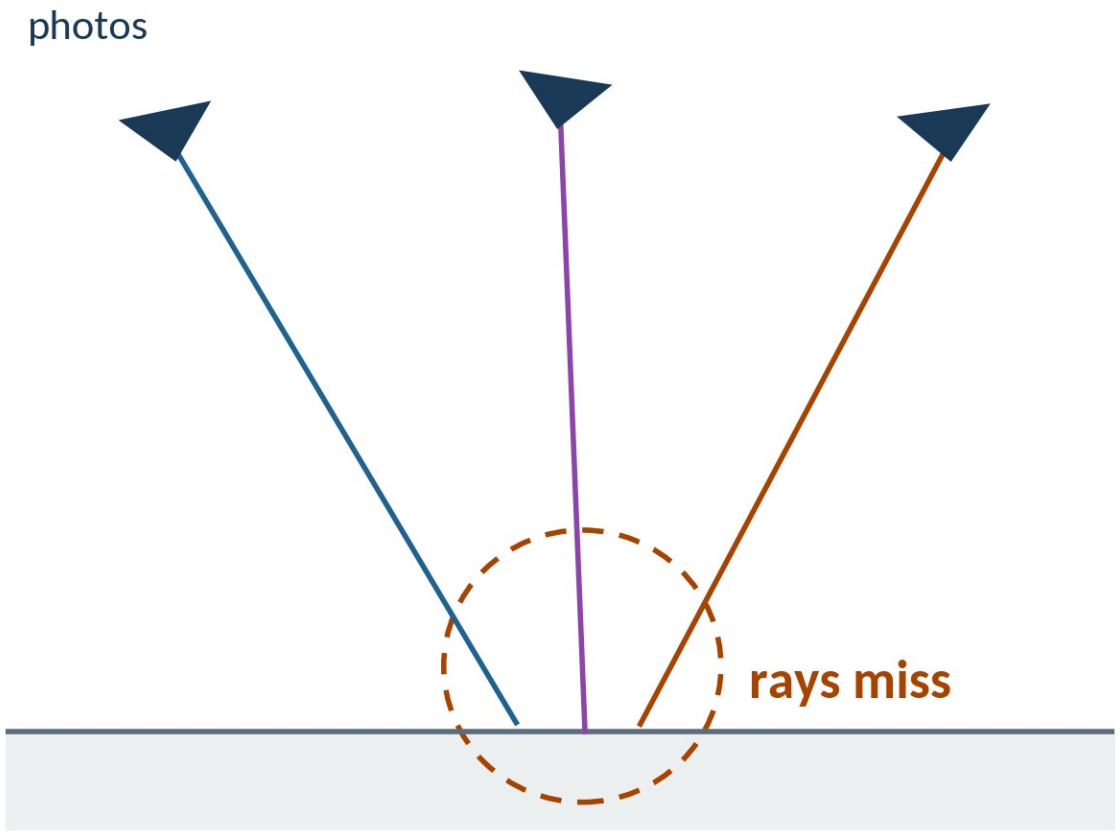


- Same feature in ≥ 3 photos (better 5+)
- Needs texture
- Scene must stand still
- Walls only from oblique photos

Further reading: Structure from motion · Feature (computer vision) · Bundle adjustment

Think first

Bundle adjustment: make the rays meet



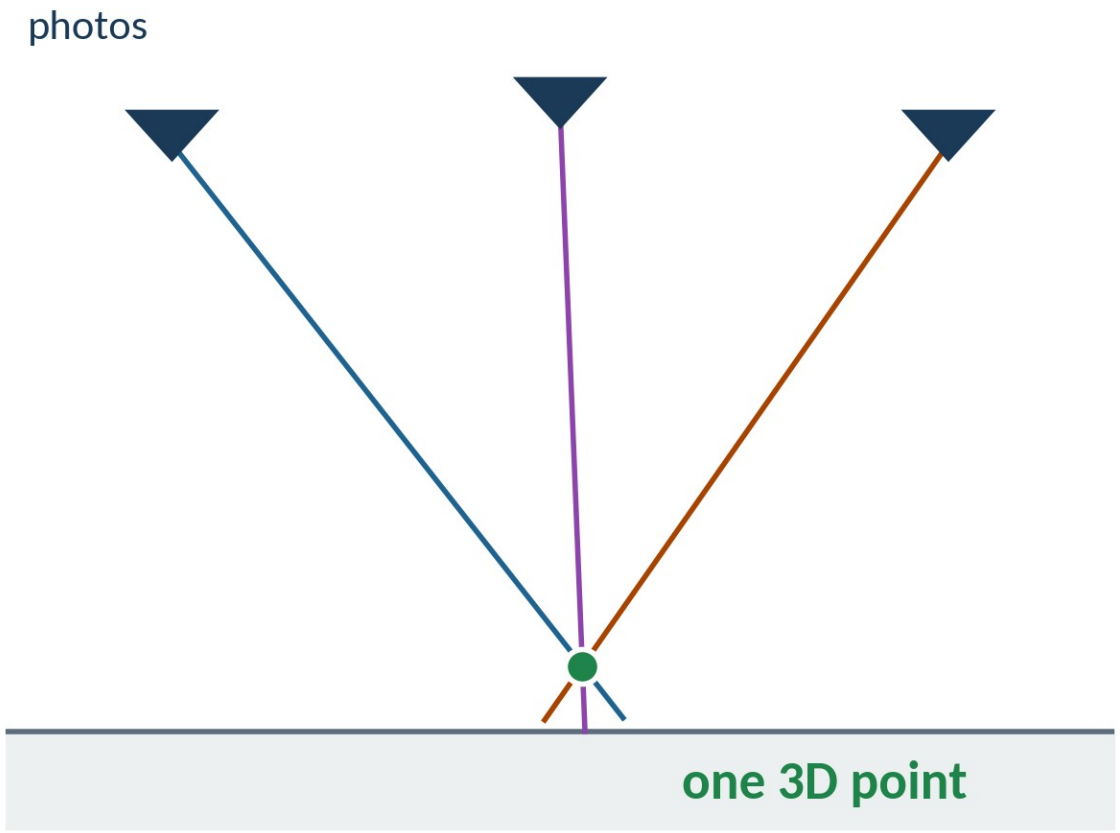
- Each photo: a bundle of rays to its tie points
- First guess (drone GNSS): rays of one point miss

Pause: what can the software change so that the rays meet?

Further reading: Bundle adjustment · Triangulation (computer vision)

Solution

Bundle adjustment: make the rays meet



- Each photo: a bundle of rays to its tie points
- First guess (drone GNSS): rays of one point miss

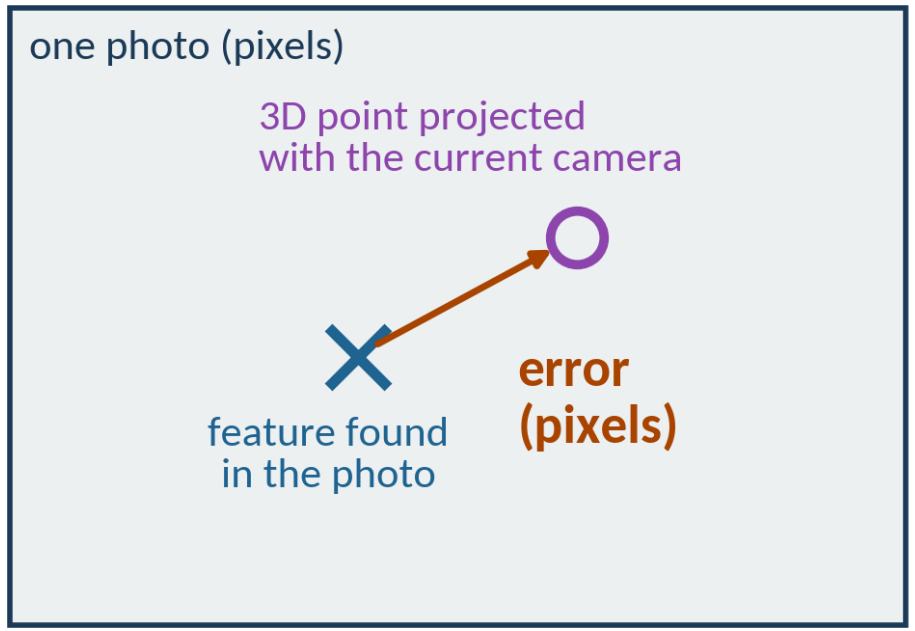
Change all at once:

- camera positions and angles (6 per photo)
- the 3D points (3 each)
- the lens model (focal length, distortion)

least squares over all photos

Further reading: Bundle adjustment · Triangulation (computer vision)

Reprojection error



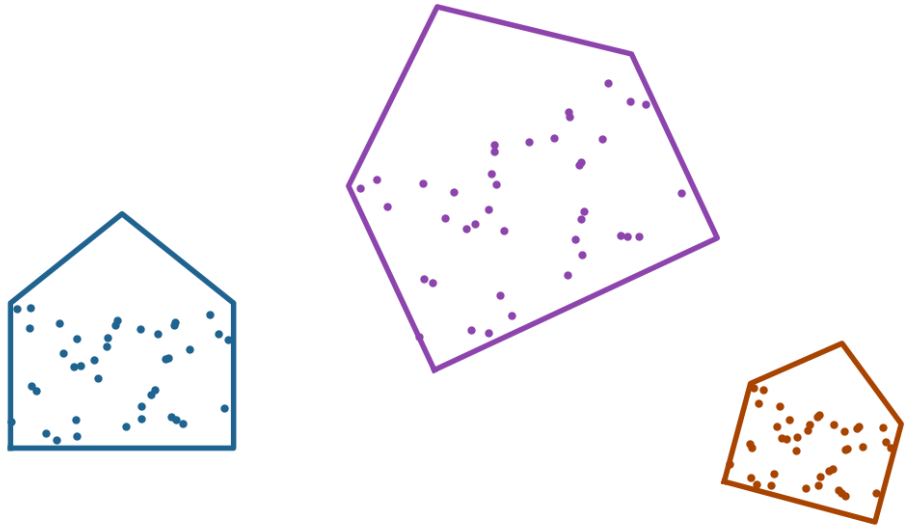
- **Minimise:** Σ (error in pixels)²
- over all points and all photos
- Green Fab Lab, RGB: 131 photos × 6 + thousands of points × 3
- **Report:** mean error < 1 px is good

Further reading: Reprojection error · Levenberg–Marquardt algorithm

Think first

A model from photos alone floats

where? how big? which way up?



- Photos fix the shape of the model
- No GNSS, no control points

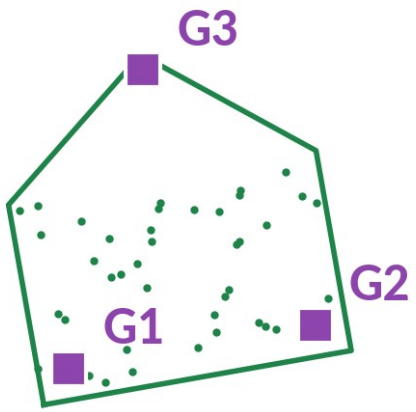
Pause: what can the photos alone not tell? how many numbers are missing?

Further reading: Similarity (geometry) · Aerial triangulation

Solution

A model from photos alone floats

fixed by ≥ 3 control points



- Photos fix the shape of the model
- No GNSS, no control points

7 numbers: 3 shift, 3 rotation, 1 scale

Fixed by:

- GCPs (≥ 3, better 5): cm
- camera positions: GNSS m, RTK/PPK cm

Check points stay out!

Further reading: Similarity (geometry) · Aerial triangulation



SfM, SLAM, multiple view geometry

Self-study

	multiple view geometry	SfM	SLAM
what	the mathematics	3D from a photo set	where am I + map
when	–	offline, after the flight	online, in real time
bundle adjustment	the method	global, all photos	local, recent frames
example	Hartley & Zisserman	OpenSfM in WebODM	robot, drone hovering

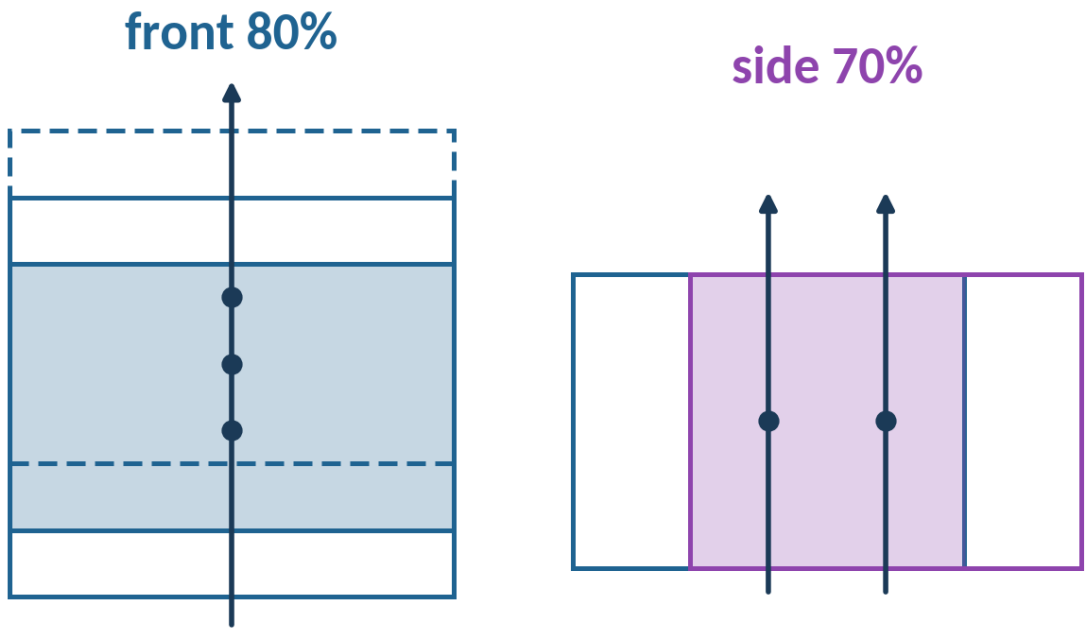
- **Photogrammetry:** the same idea as SfM, called aerial triangulation

Further reading: Structure from motion · Simultaneous localization and mapping · Epipolar geometry

Hartley, R., Zisserman, A. (2004): Multiple View Geometry in Computer Vision, 2nd ed. Cambridge University Press



Overlap



arrows: flight lines, dots: photo positions

- **Maps (2D):** 70–80 %
- **Trees, buildings:** ≈ 80–83 %
- **3D:** + cross grid, camera at 45°
- Front overlap is cheap, side overlap costs flight time

Further reading: Photogrammetry · Image stitching

Numbers: OpenDroneMap documentation, Flying tips (docs.opendronemap.org/flying)

Think first

Mavic 3M: the camera spec sheet

DJI Mavic 3M	RGB camera	multispectral camera
sensor	4/3 CMOS, 20 MP	1/2.8″ CMOS, 5 MP
image size	5280 × 3956 px	2592 × 1944 px
field of view	84° (diagonal)	73.91° (61.2° × 48.10°)
equivalent focal length	24 mm	25 mm
shortest photo interval	0.7 s (JPEG)	2 s (TIFF)
bands	RGB	G 560, R 650, RE 730, NIR 860 nm

Pause: which lines do you need for the pixel size on the ground? for the speed?

Further reading: Angle of view · 35 mm equivalent focal length

DJI Mavic 3M specifications, ag.dji.com/mavic-3-m/specs



Solution

Mavic 3M: the camera spec sheet

DJI Mavic 3M	RGB camera	multispectral camera
sensor	4/3 CMOS, 20 MP	1/2.8″ CMOS, 5 MP
image size	5280 × 3956 px	2592 × 1944 px
field of view	84° (diagonal)	73.91° (61.2° × 48.10°)
equivalent focal length	24 mm	25 mm
shortest photo interval	0.7 s (JPEG)	2 s (TIFF)
bands	RGB	G 560, R 650, RE 730, NIR 860 nm

Pixel size: image size + field of view · **speed:** photo interval

Not needed: equivalent focal length

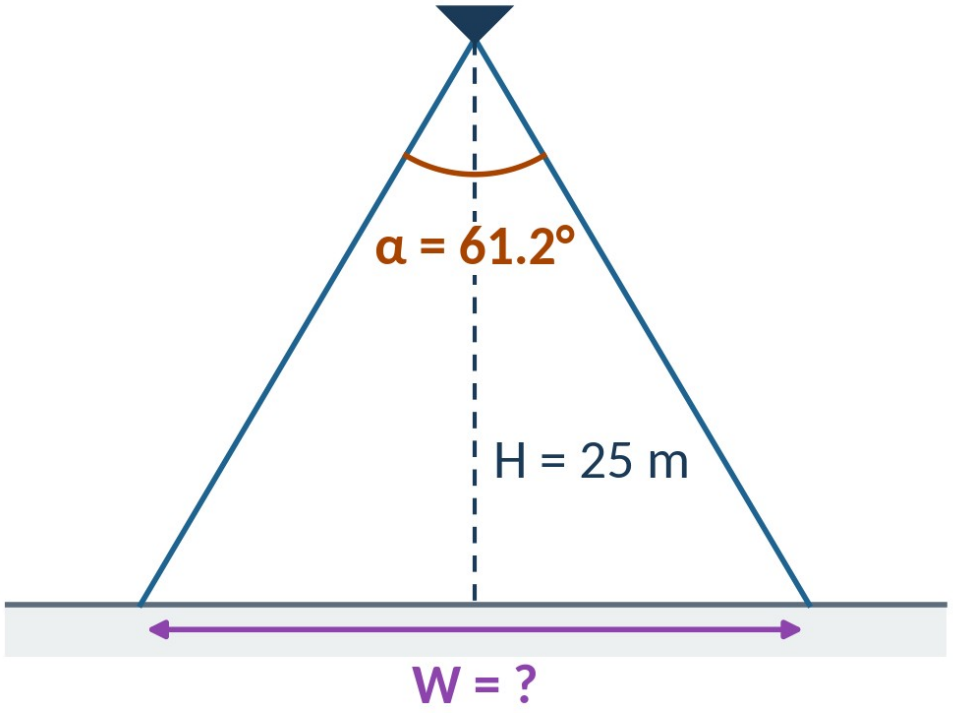
Further reading: Angle of view · 35 mm equivalent focal length

DJI Mavic 3M specifications, ag.dji.com/mavic-3-m/specs



Think first

Opening angle → footprint → pixel



Given

MS camera: 61.2° across 2592 px

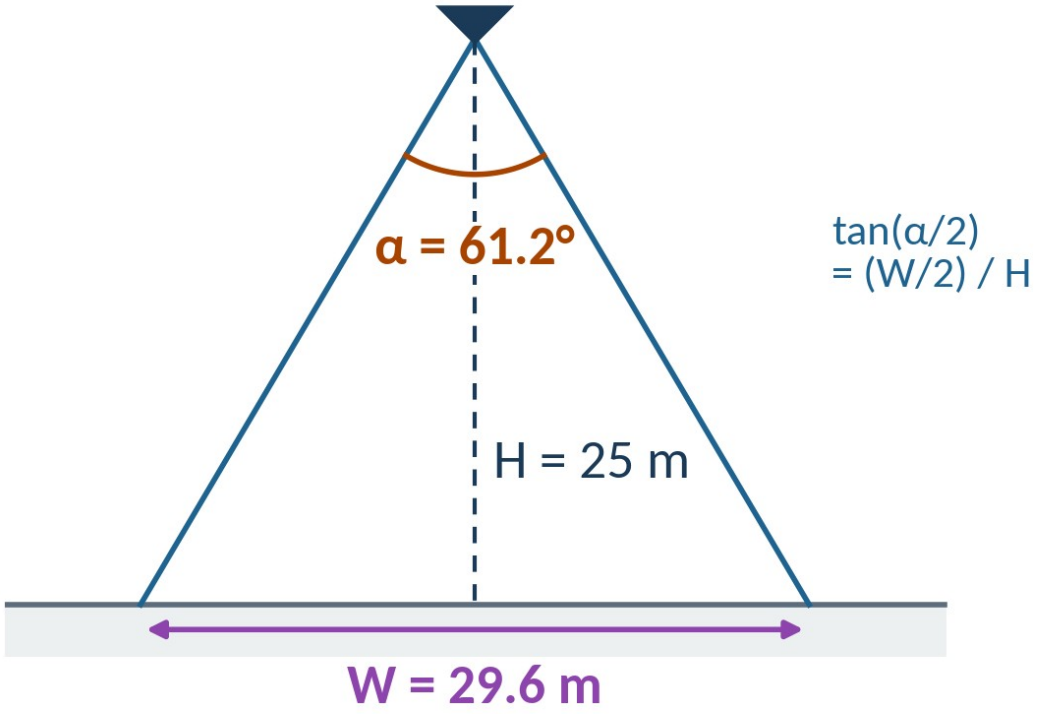
flying height H = 25 m

Pause: how wide is one photo on the ground? how large is one pixel?

Further reading: Angle of view · Ground sample distance

Solution

Opening angle → footprint → pixel



tan(α/2)
= (W/2) / H

Given

MS camera: 61.2° across 2592 px

flying height H = 25 m

1 **W = 2 · H · tan(α/2)**

= 2 · 25 m · tan(30.6°) = 29.6 m

2 **GSD = W / pixels**

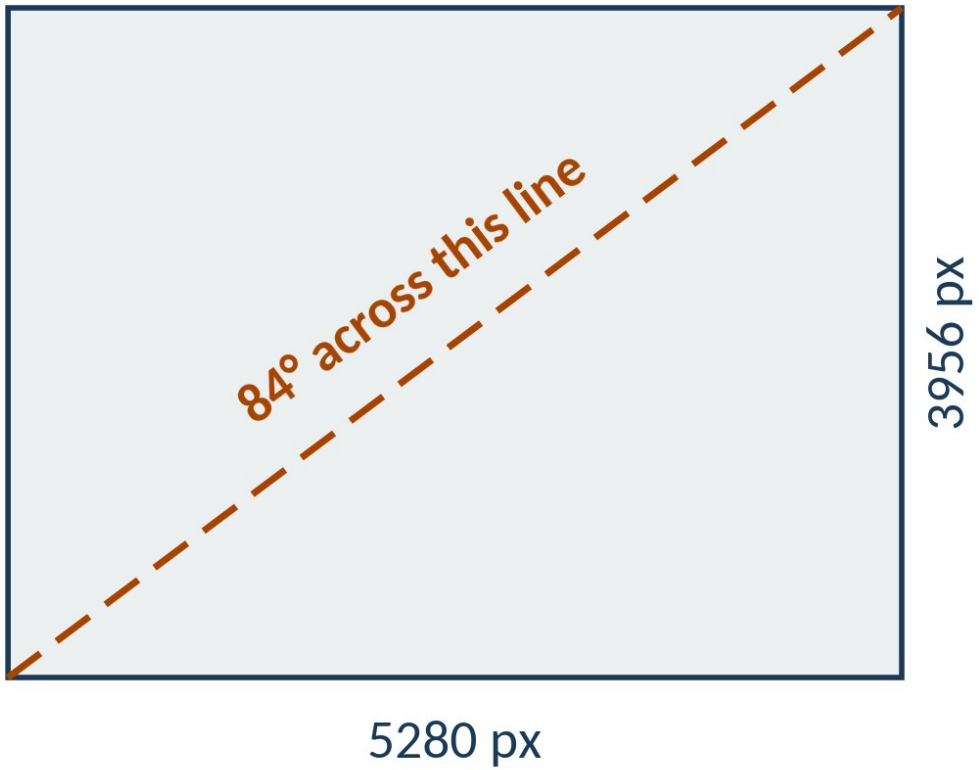
= 29.6 m / 2592 = 1.14 cm

Rule: GSD in cm ≈ H in m / 21.9

Further reading: Angle of view · Ground sample distance

Think first

RGB: the angle is across the diagonal



Given

RGB camera: 84° across the diagonal

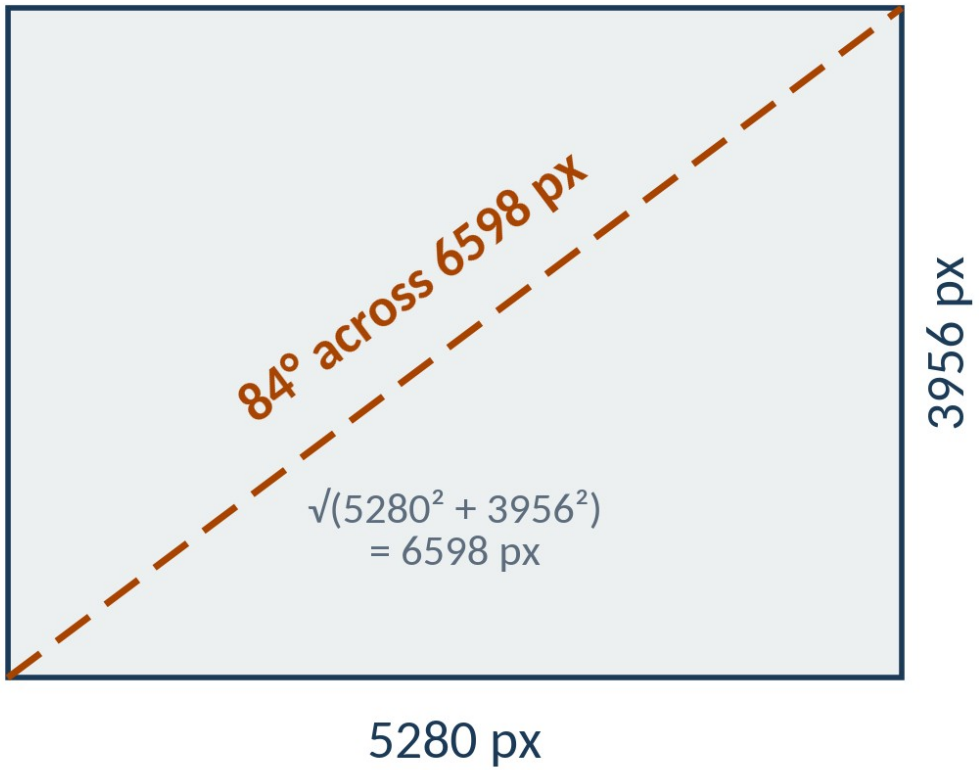
5280 × 3956 px, H = 25 m

Pause: the pixels are counted along x and y, the angle along the diagonal: how do you get the pixel size?

Further reading: Angle of view

Solution

RGB: the angle is across the diagonal



Given

RGB camera: 84° across the diagonal

5280 × 3956 px, H = 25 m

1 diagonal: $\sqrt{(5280^2 + 3956^2)} = 6598$ px

2 $D = 2 \cdot 25\ m \cdot \tan(42°) = 45.0$ m

3 **GSD = 45.0 m / 6598 = 0.68 cm**

photo: 36.0 × 27.0 m

Not: 84° × 5280/6598 = 67.2°

Further reading: Angle of view

Is the spec sheet consistent?

Self-study

check	computed	DJI states	agrees?
RGB: 24 mm equivalent → diagonal	84.1°	84°	yes
MS: 25 mm equivalent → diagonal	81.7°	73.91°	no
MS: 61.2° × 48.10° → diagonal	73.1°	73.91°	≈ 1 %

- **Rule:** use the angle that belongs to your pixel count
- Equivalent focal length: for photographers, not for mapping

DJI Mavic 3M specifications; full frame = 36 × 24 mm



Pixel size on the ground (GSD)

GSD = H × (width of the view on the ground per metre) / pixels

height	RGB pixel	MS pixel	RGB photo
25 m	0.7 cm	1.1 cm	36 × 27 m
40 m	1.1 cm	1.8 cm	58 × 43 m
60 m	1.6 cm	2.7 cm	86 × 65 m
120 m	3.3 cm	5.5 cm	173 × 130 m

Mavic 3M, GSD in cm: RGB ≈ H / 36.6, multispectral ≈ H / 21.9 (H in m)

Further reading: Ground sample distance

Camera: DJI Mavic 3M specifications (ag.dji.com/mavic-3-m/specs)



Sharp photos

blur = speed × exposure time (keep < ½ pixel)

speed	exposure	blur	at 25 m (RGB 0.68 cm)
2.2 m/s	1/1000 s	2.2 mm	0.3 px
7.7 m/s	1/1000 s	7.7 mm	1.1 px
7.7 m/s	1/250 s	30.8 mm	4.5 px

- Low light → longer exposure → fly slower
- **RGB: mechanical shutter** · MS: electronic

Further reading: Motion blur · Shutter speed · Rolling shutter



Light, weather, scene

- **Even light:** overcast is fine, broken clouds are not
- **November sun:** ≈ 23° at noon → shadows 2.4 × object height
- **Wind:** moving trees, blurred photos
- **Multispectral:** light sensor + reflectance panel before and after
- **Scene:** no water, glass, moving people or cars

Further reading: Solar zenith angle · Multispectral imaging



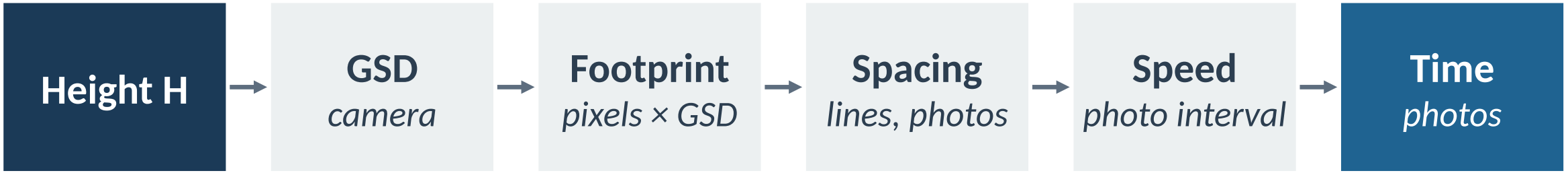
3 Plan

From flying height to flight lines

Photos → map | Control | Photos | **Plan** | Results



From flying height to flight time

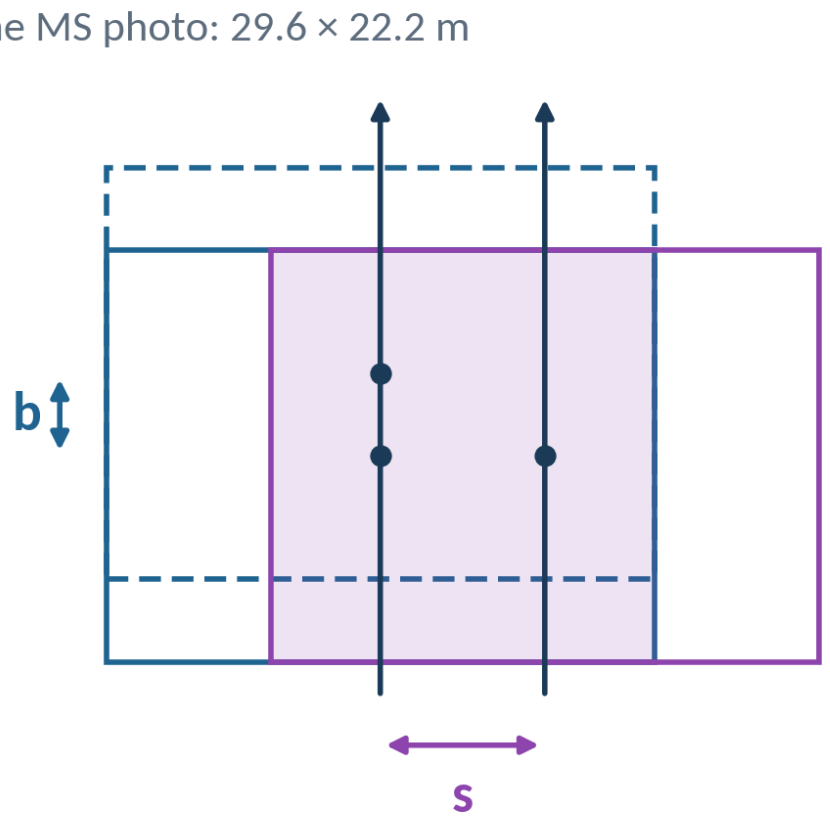


- **Line spacing** = footprint across × (1 − side overlap)
- **Photo spacing** = footprint along × (1 − front overlap)
- **Speed** = photo spacing / photo interval (RGB 0.7 s, MS 2 s)
- **Lower**: finer pixels, more lines, longer flight

Further reading: Ground sample distance · Photogrammetry

Think first

Line spacing



Given

MS photo across the line: 29.6 m

side overlap 70 %

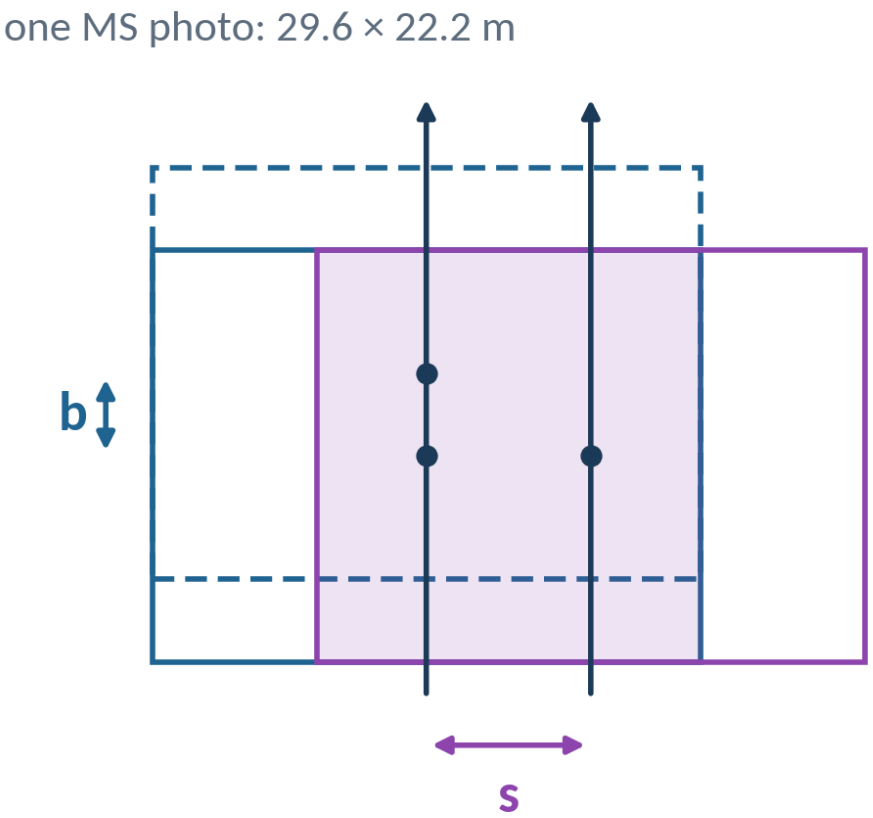
area ≈ 48 m across the lines

Pause: how far apart are the flight lines (s)? how many lines?

Further reading: Photogrammetry

Solution

Line spacing



Given

MS photo across the line: 29.6 m

side overlap 70 %

area ≈ 48 m across the lines

1 **s = across · (1 – side overlap)**

= 29.6 m · 0.3 = 8.9 m

2 lines = ⌈48 / 8.9⌉ = 6

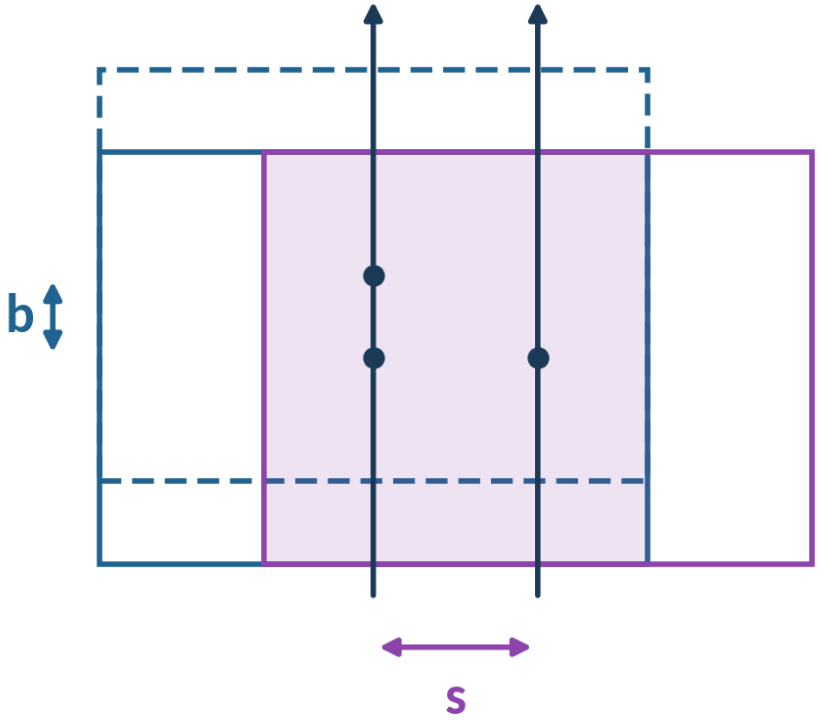
RGB alone: 10.8 m → 5 lines

Further reading: Photogrammetry

Think first

Photo spacing and speed

one MS photo: 29.6 × 22.2 m



Given

MS photo along the line: 22.2 m

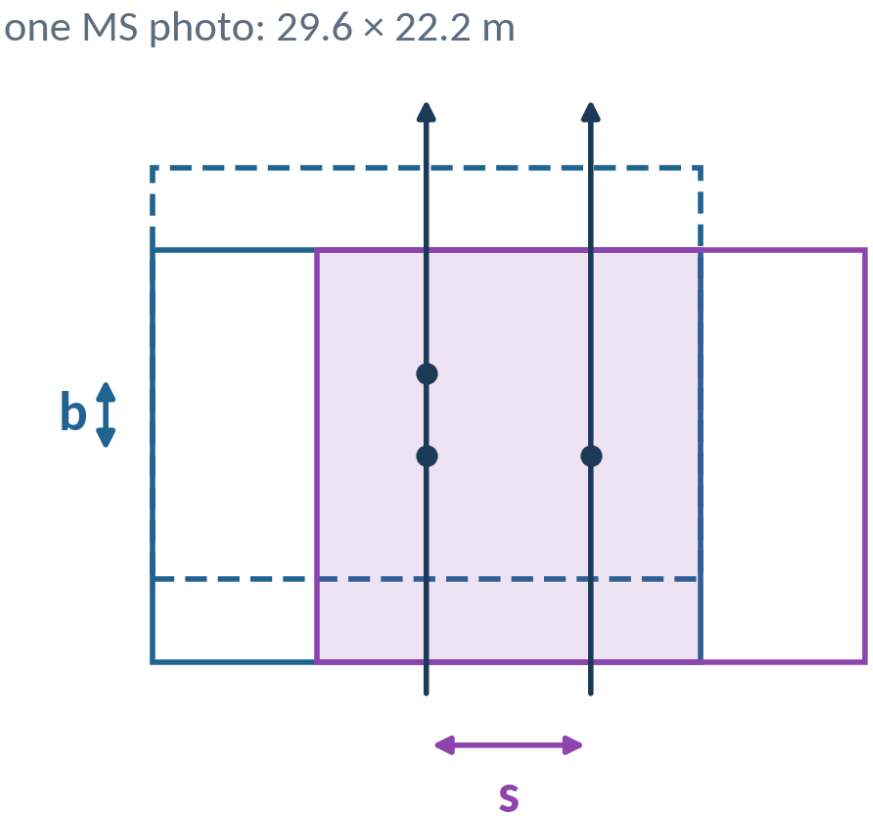
front overlap 80 %

MS: one photo every 2 s

Pause: how far apart are the photos (b)? how fast may the drone fly?

Solution

Photo spacing and speed



Given

MS photo along the line: 22.2 m

front overlap 80 %

MS: one photo every 2 s

1 **b = along · (1 − front overlap)**

= 22.2 m · 0.2 = 4.4 m

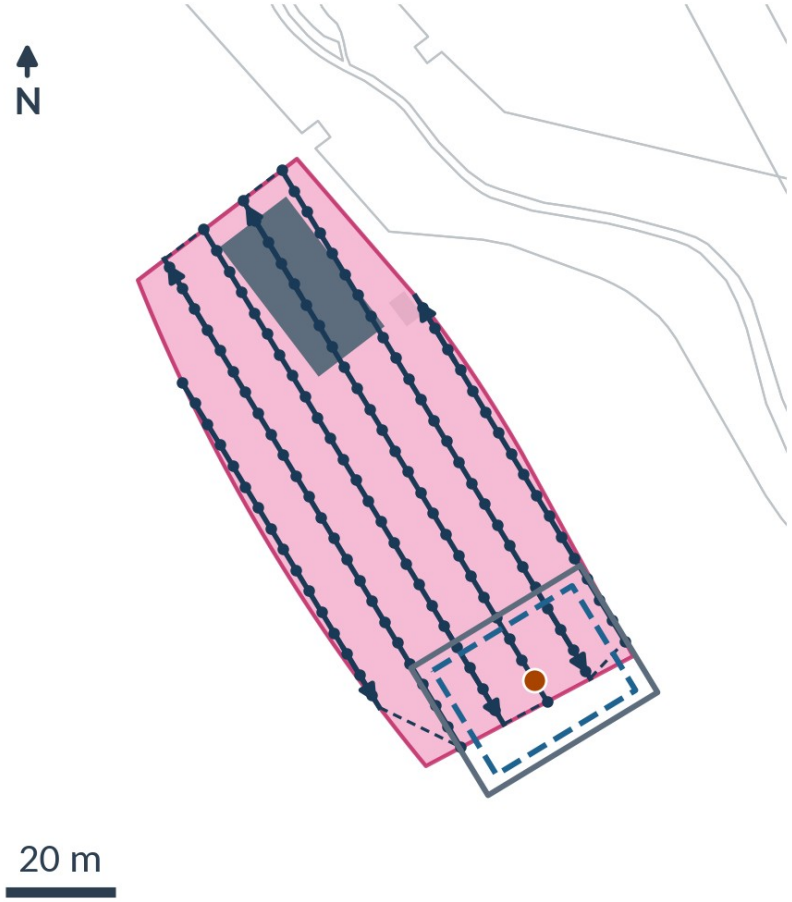
2 **v = b / interval = 4.4 m / 2 s = 2.2 m/s**

RGB alone: 5.4 m / 0.7 s = 7.7 m/s

→ the MS camera decides

Think first

Flight time and number of photos



Given

6 lines, path 618 m (lines + turns)

v = 2.2 m/s, ≈ 10 s per turn

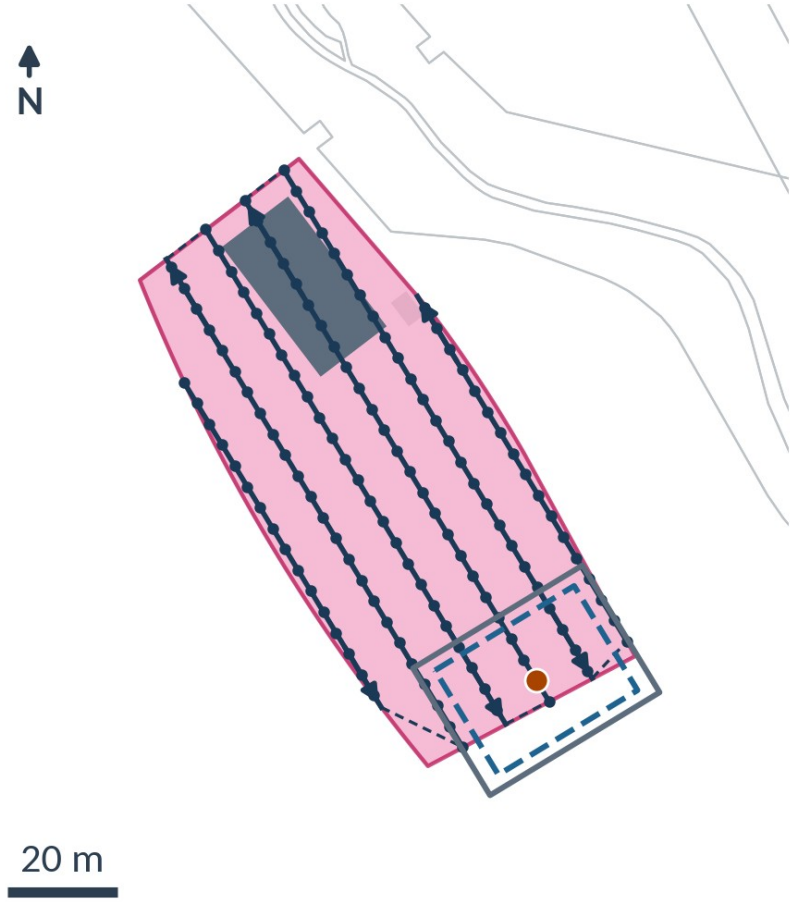
photo every 4.4 m

Pause: how long is the flight? how many photo positions and image files?

Further reading: Photogrammetry

Solution

Flight time and number of photos



Given

6 lines, path 618 m (lines + turns)

v = 2.2 m/s, ≈ 10 s per turn

photo every 4.4 m

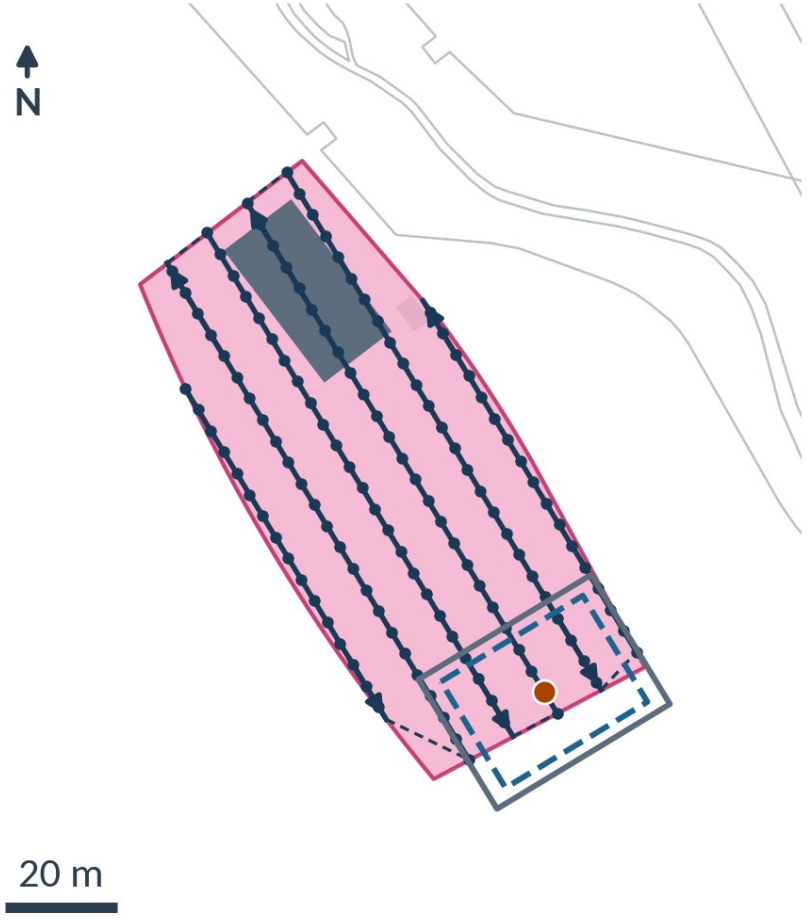
1 t = 618 m / 2.2 m/s + 5 × 10 s
= 279 s + 50 s ≈ 5.5 min

2 per line ⌊length / b⌋ + 1 → 131 positions

3 × 5 images (RGB + 4 bands) ≈ 655 files

Further reading: Photogrammetry

The Green Fab Lab plan



all cameras, 25 m	
pixel RGB / MS	0.7 / 1.1 cm
lines, spacing	6 × 8.9 m
photo every	4.4 m
speed	2.2 m/s
photo positions	131
flight time	≈ 5 min

Map: ALKIS (simplified WFS) and height benchmarks: Geobasis NRW, dl-de/zero-2-0; plan: R. Becker (common.mission); camera: DJI Mavic 3M specs

All cameras: the multispectral camera decides

at 25 m	RGB	multispectral
pixel	0.68 cm	1.14 cm
photo on the ground	36 × 27 m	30 × 22 m
line spacing (70 %)	10.8 m	8.9 m
photo spacing (80 %)	5.4 m	4.4 m
photo interval	0.7 s	2 s
max. speed	7.7 m/s	2.2 m/s

Pause: what changes if we fly at 40 m?



Before take-off

Rules

Registered operator, certified pilot

Visual line of sight, ≤ 120 m

Not over uninvolved people

Geo-zones: dipul.de

On site

Public park: watch the paths

Wind, rain, light

Batteries, home point

RTK fix before take-off

Further reading: Regulation of unmanned aerial vehicles

Rules: Regulation (EU) 2019/947 (open category); geo-zones: dipul.de. The pilot in command decides.



Plan in Python, fly with DJI Pilot 2



- You compute the numbers; the app flies the lines
- Same height and overlap in both
- **Homework:** your plan for the Green Fab Lab

Further reading: Keyhole Markup Language

Summary

- GCPs fix the model, check points test it – never mix them
- Drone RTK: GCPs optional, check points still needed
- Every point in ≥ 3–5 photos: 70–80 % overlap, texture, a still scene
- Height sets pixel size, lines, photos and time
- **All cameras: the multispectral camera decides**



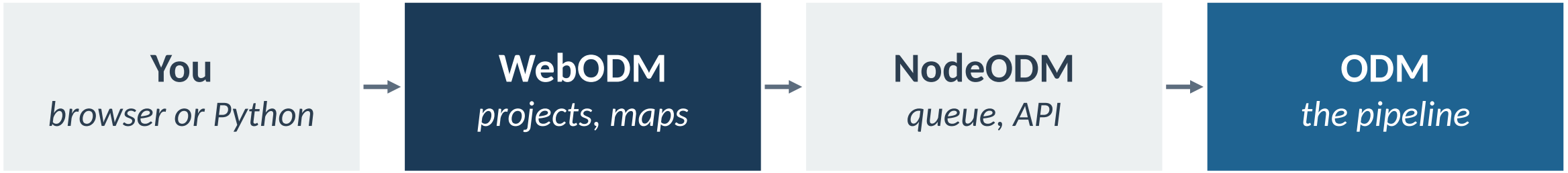
4 Results

WebODM, orthophoto, DSM, accuracy

Photos → map | Control | Photos | Plan | **Results**



WebODM, NodeODM, ODM



- **ODM:** OpenSfM → OpenMVS → DSM, DTM, orthophoto, report
- **WebODM:** upload, options, GCP interface, maps, downloads
- **Open source:** runs on a laptop, a lab server, the cloud

Further reading: OpenDroneMap docs · WebODM docs · Structure from motion

OpenDroneMap, WebODM, NodeODM: opendronemap.org (AGPL-3.0)

One task: what goes in, what comes out

In

Photos (GNSS in EXIF)

`gcp_list.txt` ground control

`geo.txt` camera positions (PPK)

Options

→

Out

Orthophoto (GeoTIFF)

DSM, DTM (GeoTIFF)

Point cloud (LAZ), 3D model

Quality report (PDF)

Same photos, other control or options = **a new task**

Further reading: ODM outputs · Image geolocation files · Orthophoto



Processing options that matter

option	what it does	our choice
dsm, dtm	surface and terrain model	on
orthophoto-, dem-resolution	cm per pixel (≥ GSD)	≈ GSD
auto-boundary	cut to the photographed area	on
gps-accuracy	trust in the photo positions	3 m / 0.05 m RTK
force-gps	photo positions + GCPs	RTK photos + GCPs
fast-orthophoto	quick look, no dense cloud	field check only

Comparison: change only the control, keep all other options

Further reading: ODM options · Digital elevation model



Our flights as tasks

photo set	camera positions	without GCPs	with 5 GCPs
F2c	drone GNSS (EXIF)	m?	cm?
F2c + PPK	geo.txt from Emlid Studio	cm?	cm?
F2b	RTK from our base (B1)	cm + shift?	cm?
F2a	RTK from SAPOS	cm?	cm?

- Same mission, same options – **only the positions change**
- **Check points C1–C3:** never in a task

Further reading: Real-time kinematic positioning · Emlid Studio: PPK for drones



Clicking the GCPs

- **≥ 5 GCPs × ≥ 3 photos** = at least 15 lines
- **Click the centre** of the checkerboard, at full zoom
- Prefer photos with the panel near the image centre
- One height system for all points (DHHN2016)
- **Check points:** never clicked

Further reading: ODM: ground control points · Ground control point

Rules: OpenDroneMap documentation, Ground Control Points (docs.opendronemap.org/gcp)



Processing a task: how to

- **Token** from your course account
- **Options** as name/value list
- **Status 40:** download the assets

```
import json, requests
URL = "https://odm.eolab.de/api"
r = requests.post(URL + "/token-auth/",
    data={"username": USER, "password": PW})
H = {"Authorization":
     "JWT " + r.json()["token"]}
opts = [{"name": k, "value": True} for k in
        ("dsm", "dtm", "auto-boundary")]
files = [("images", open(f, "rb"))
         for f in photos + ["gcp_list.txt"]]
task = requests.post(
    f"{URL}/projects/{pid}/tasks/", headers=H,
    files=files, data={"name": "F2a_gcp",
    "options": json.dumps(opts)}).json()
# poll .../tasks/<id>/ until status 40, then
# GET .../tasks/<id>/download/orthophoto.tif
```

Further reading: WebODM API quickstart · WebODM task API



The quality report

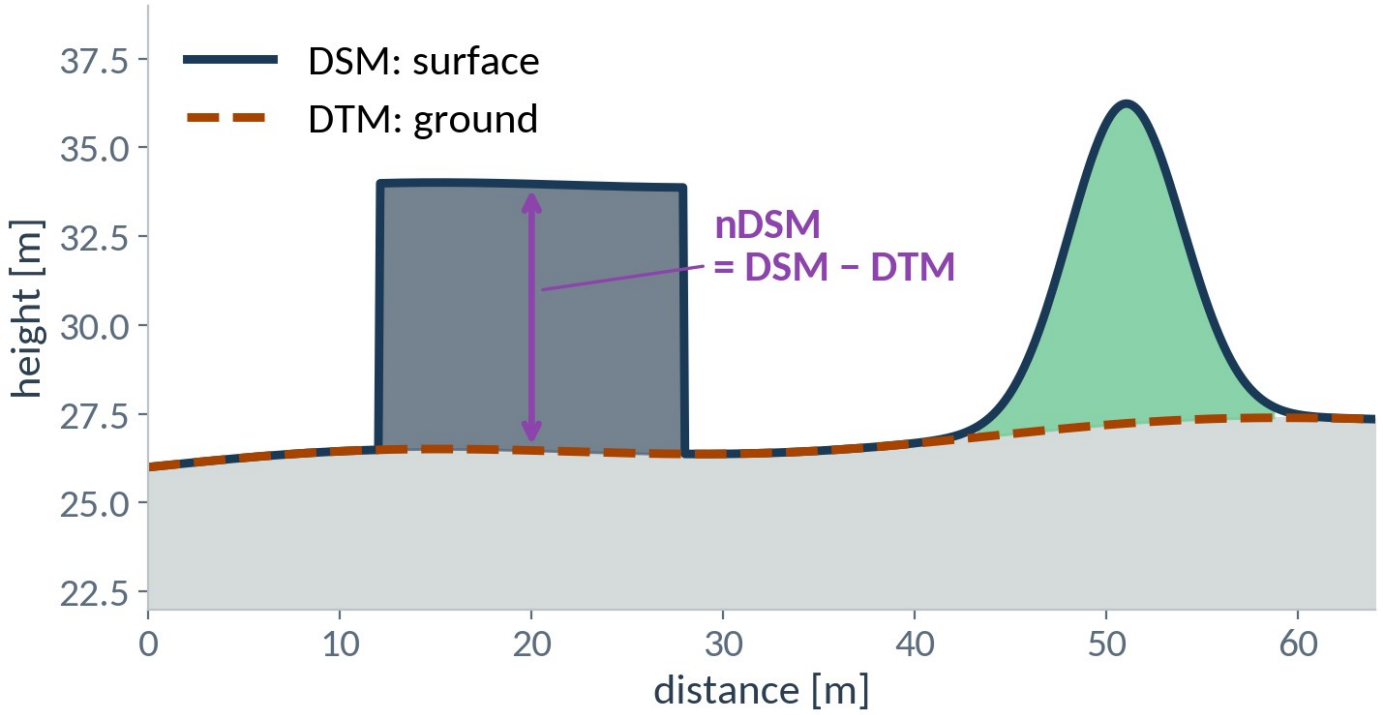
look at	good sign
photos reconstructed	all (or nearly all)
reprojection error	< 1 px
overlap map	≥ 5 photos over the area
GCP residuals	cm, no outlier

- **GCP residuals fit, they do not test**
- **Accuracy comes from the check points**

Further reading: ODM outputs · Bundle adjustment



Orthophoto, DSM, DTM



- **DSM:** roofs, crowns
- **DTM:** ground (filtered)
- **nDSM:** object heights
- **Ortho:** photos on the DSM, true scale

Edges of roofs and trees: seams and leaning walls

Further reading: Digital elevation model · Orthophoto · DGM1 / DOM1 NRW

Figure: R. Becker (make_slides.py), CC BY-SA 4.0, schematic

Accuracy on check points

- **Per point:** ΔE, ΔN, ΔH = map − RTK
- **Horizontal:** RMSE = $\sqrt{(\text{mean } \Delta E^2 + \text{mean } \Delta N^2)}$
- **Vertical:** RMSE = $\sqrt{(\text{mean } \Delta H^2)}$
- **Mean ≫ spread:** a shift, not noise

ASPRS: ≥ 30 check points for an accuracy statement

3 check points: an indication only

E, N: panel centre on the ortho · **H:** DSM at the RTK point

Further reading: ASPRS Positional Accuracy Standards · Root-mean-square deviation

ASPRS Positional Accuracy Standards for Digital Geospatial Data, Edition 2 (2023/2024)



Think first

Which error do you expect?

processing	horizontal	vertical
drone GNSS only	?	?
+ PPK or RTK (SAPOS)	?	?
RTK from our base	?	?
+ 5 GCPs	?	?

Pause: fill in the table – then compare with your own results

Further reading: Accuracy and precision



Solution

Which error do you expect?

processing	horizontal	vertical
drone GNSS only	m	several m
+ PPK or RTK (SAPOS)	2–5 cm	3–10 cm
RTK from our base	cm + shift of B1	cm + shift
+ 5 GCPs	1–2 GSD	2–3 GSD

Orders of magnitude – your check points give the real values

Further reading: Accuracy and precision



A 45 m error is not a bad flight

pattern of Δ	likely cause
ΔH ≈ 45 m at every point	ellipsoid h vs DHHN2016 H
ΔH ≈ 1.5–2 m	pole or antenna height
same few cm everywhere	base coordinate, datum
one point far off	typo, moved panel, name
small, different at each point	the real accuracy

First the pattern, then the number

Further reading: Geoid · GCG2016 (BKG)



Drone DSM and the ALS models

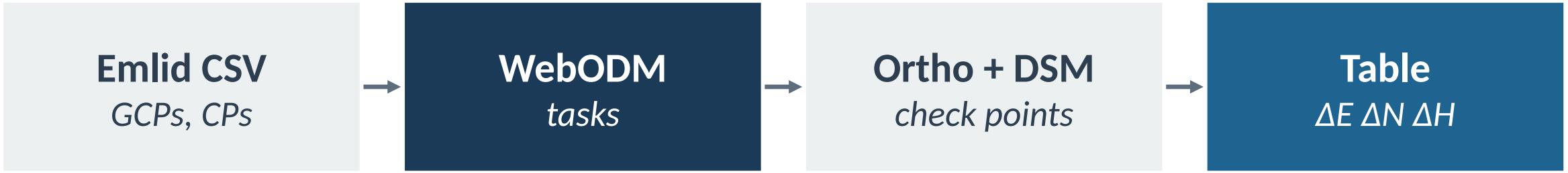
- **Drone DSM – DGM1** = object heights (roof, trees)
- **Drone DSM – DOM1** on paving = height check, no field work
- **Differences on plants:** change since the ALS flight
- **First:** same 1 m grid, same height system

Further reading: Lidar · DGM1 / DOM1 NRW

DGM1, DOM1: Geobasis NRW, dl-de/zero-2-0 (geo1040)



Exercise geo1263: your accuracy table



- One photo set per group, with and without GCPs
- Errors on C1–C3 for all tasks of the course
- **Stretch:** drone DSM vs DOM1

Summary: results

- One task = one photo set + one control + one set of options
- Change only the control when you compare
- GCP residuals fit the model; check points test it
- Look at the pattern first: 45 m is a height system
- **Cheapest method that meets the purpose, proven with check points**



Sources and further reading (1/4)

From photos to a map: Photogrammetry · Structure from motion · Bundle adjustment · Orthophoto · Digital elevation model

Ground control and check points: Ground control point · Accuracy and precision · Root mean square deviation

How many, and where?: Ground control point · Bundle adjustment

Georeferencing: two questions: Real-time kinematic positioning · Ground control point · Orthophoto

What must my map fit?: Image registration · Accuracy and precision

Our choice at the Green Fab Lab: Real-time kinematic positioning

Targets you can find in the photos: Fiducial marker · Ground sample distance

Our base mark at the Green Fab Lab: Survey marker · Tribrach

Structure from motion: Structure from motion · Feature (computer vision) · Bundle adjustment

Bundle adjustment: make the rays meet: Bundle adjustment · Triangulation (computer vision)

Reprojection error: Reprojection error · Levenberg–Marquardt algorithm

A model from photos alone floats: Similarity (geometry) · Aerial triangulation



Sources and further reading (2/4)

SfM, SLAM, multiple view geometry: Structure from motion · Simultaneous localization and mapping · Epipolar geometry

Overlap: Photogrammetry · Image stitching

Mavic 3M: the camera spec sheet: Angle of view · 35 mm equivalent focal length

Opening angle → footprint → pixel: Angle of view · Ground sample distance

RGB: the angle is across the diagonal: Angle of view

Pixel size on the ground (GSD): Ground sample distance

Sharp photos: Motion blur · Shutter speed · Rolling shutter

Light, weather, scene: Solar zenith angle · Multispectral imaging

From flying height to flight time: Ground sample distance · Photogrammetry

Line spacing: Photogrammetry

Flight time and number of photos: Photogrammetry

Before take-off: Regulation of unmanned aerial vehicles



Sources and further reading (3/4)

Plan in Python, fly with DJI Pilot 2: Keyhole Markup Language

WebODM, NodeODM, ODM: OpenDroneMap docs · WebODM docs · Structure from motion

One task: what goes in, what comes out: ODM outputs · Image geolocation files · Orthophoto

Processing options that matter: ODM options · Digital elevation model

Our flights as tasks: Real-time kinematic positioning · Emlid Studio: PPK for drones

Clicking the GCPs: ODM: ground control points · Ground control point

Processing a task: how to: WebODM API quickstart · WebODM task API

The quality report: ODM outputs · Bundle adjustment

Orthophoto, DSM, DTM: Digital elevation model · Orthophoto · DGM1 / DOM1 NRW

Accuracy on check points: ASPRS Positional Accuracy Standards · Root-mean-square deviation

Which error do you expect?: Accuracy and precision

A 45 m error is not a bad flight: Geoid · GCG2016 (BKG)



Sources and further reading (4/4)

Drone DSM and the ALS models: Lidar · DGM1 / DOM1 NRW

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